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COHORT C

```
MariaDB [(none)]> create database Projects;  
Query OK, 1 row affected (0.004 sec)
```

```
MariaDB [(none)]> use Projects;  
Database changed
```

```
MariaDB [(none)]> use Projects;  
Database changed
```

```
MariaDB [Projects]> create table Employee(empNo varchar (4)  
primary key, fName varchar(30), lName varchar(30), address  
varchar(30), DOB date, sex varchar (1) check(sex in ('M' , 'F')),  
position varchar(18) check (position in ('Manager', 'Team  
Leader' , 'Analyst' , 'Software Developer')), deptNo varchar  
(4));  
Query OK, 0 rows affected (0.031 sec)
```

```
MariaDB [Projects]> create table Department(deptNo varchar (4)  
primary key, deptName varchar (15), mgrEmpNo varchar(4), foreign  
key (mgrEmpNo) references Employee(empNo));  
Query OK, 0 rows affected (0.026 sec)
```

```
MariaDB [Projects]> create table WorksOn(empNo varchar(4),  
projNo varchar(4), dateWorked date, hoursWorked int check  
(hoursWorked between 0 and 40), primary key (empNo, projNo,  
dateWorked), foreign key (empNo) references Employee(empNo),  
foreign key (projNo) references Project(projNo));  
Query OK, 0 rows affected (0.031 sec)
```

```
MariaDB [Projects]> show tables;
```

Tables_in_Projects
Department
Employee
Project
WorksOn

```
4 rows in set (0.001 sec)
```

```
MariaDB [Projects]>
```

```
MariaDB [Projects]> create view FemaleManagedProjects as select  
k.projNo, k.projName, k.deptNo from Project k, Department c,  
Employee w where k.deptNo= c.deptNo and c.mgrEmpNo = w.empNo and  
w.sex = 'F' order by k.projNo;
```

```
Query OK, 0 rows affected (0.024 sec)
```

```
MariaDB [Projects]> create view EmpProjectHours as select  
k.empNo, k.fName, k.lName, w.projName, c.hoursWorked from  
Employee k, Project w, WorksOn c where c.empNo = k.empNo and  
c.projNo = w.projNo;
```

```
Query OK, 0 rows affected (0.023 sec)
```

```
MariaDB [Projects]> create view EmpProject(empNo, projNo,  
totalHours) as select w.empNo, w.projNo, sum(hoursWorked) from  
Employee e, Project p, WorksOn w where e.empNo = w.empNo and  
p.projNo = w.projNo group by w.empNo, w.projNo;
```

```
Query OK, 0 rows affected (0.020 sec)
```

```
MariaDB [Projects]> insert into Employee values ('E1', 'Kevin',  
'Wilson', 'Oxford St, Osu', '1985-04-10', 'M', 'Manager', NULL);
```

```
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [Projects]> insert into Employee values ('E2',  
'Emmanuel', 'Mensah', 'Ring Rd Central', '1990-07-22', 'M',  
'Manager', NULL);
```

```
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [Projects]> insert into Employee values ('E3', 'Kelvin',  
'James', 'Liberation Rd', '1988-01-15', 'M', 'Software  
Developer', NULL);
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into Employee values ('E4',  
'Michael', 'Prince', 'Spintex Rd', '1992-03-30', 'M', 'Analyst',  
NULL);
```

Query OK, 1 row affected (0.002 sec)

```
MariaDB [Projects]> insert into Employee values ('E5', 'Peter',  
'Chech', 'Independence Ave', '1987-11-05', 'M', 'Team Leader',  
NULL);
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into Employee values ('E6',  
'Deborah', 'Mills', 'High St, Takoradi', '1991-06-18', 'F',  
'Analyst', NULL);
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into Employee values ('E7', 'Babara',  
'Cole', 'Kwame Nkrumah Ave, Kumasi', '1986-09-12', 'F',  
'Software Developer', NULL);
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into Employee values ('E8', 'James',  
'Johnson', 'Cantonments Rd', '1993-02-25', 'M', 'Team Leader',  
NULL);
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into Department values ('D1', 'IT',  
'E1');
```

Query OK, 1 row affected (0.003 sec)

```
MariaDB [Projects]> insert into Department values ('D2',  
'Research', 'E2');
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into Department values ('D3',  
'Finance', 'E3');
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into Department values ('D4',  
'Marketing', 'E4');
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into Department values ('D5',  
'Operations', 'E5');
```

```
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [Projects]> insert into Department values ('D6', 'HR',  
'E6');
```

```
Query OK, 1 row affected (0.002 sec)
```

```
MariaDB [Projects]> insert into Department values ('D7',  
'Sales', 'E7');
```

```
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [Projects]> insert into Department values ('D8',  
'Support', 'E8');
```

```
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [Projects]> update employee set deptno = 'D1' where  
empno = 'E1';
```

```
Query OK, 1 row affected (0.005 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [Projects]> update employee set deptno = 'D2' where  
empno = 'E2';
```

```
Query OK, 1 row affected (0.002 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [Projects]> update employee set deptno = 'D3' where  
empno = 'E3';
```

```
Query OK, 1 row affected (0.001 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [Projects]> update employee set deptno = 'D4' where  
empno = 'E4';
```

```
Query OK, 1 row affected (0.006 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [Projects]> update employee set deptno = 'D5' where  
empno = 'E5';
```

```
Query OK, 1 row affected (0.001 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [Projects]> update employee set deptno = 'D6' where  
empno = 'E6';
```

```
Query OK, 1 row affected (0.001 sec)
```

Rows matched: 1 Changed: 1 Warnings: 0

```
MariaDB [Projects]> update employee set deptno = 'D7' where  
empno = 'E7';
```

Query OK, 1 row affected (0.002 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
MariaDB [Projects]> update employee set deptno = 'D8' where  
empno = 'E8';
```

Query OK, 1 row affected (0.001 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
MariaDB [Projects]> insert into project values ('SCCS',  
'System', 'D1');
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into project values ('P2', 'WebApp',  
'D2');
```

Query OK, 1 row affected (0.002 sec)

```
MariaDB [Projects]> insert into project values ('P3', 'Banking',  
'D3');
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into project values ('P4',  
'Consultancy', 'D6');
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into project values ('P5',  
'Database', 'D7');
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into worksOn values ('E1', 'SCCS',  
'2026-06-01', 10);
```

Query OK, 1 row affected (0.003 sec)

```
MariaDB [Projects]> insert into workson values ('E1', 'SCCS',  
'2026-06-02', 8);
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into worksOn values ('E2', 'P2',  
'2026-06-01', 12);
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into worksOn values ('E3', 'P3',  
'2026-06-01', 15);
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into worksOn values ('E4', 'SCCS',  
'2026-06-03', 6);
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into worksOn values ('E6', 'P4',  
'2026-06-01', 20);
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> insert into worksOn values ('E7', 'P5',  
'2026-06-01', 18);
```

Query OK, 1 row affected (0.001 sec)

```
MariaDB [Projects]> select * from EmpProject;
```

empNo	projNo	totalHours
E1	SCCS	18
E2	P2	12
E3	P3	15
E4	SCCS	6
E6	P4	20
E7	P5	18

6 rows in set (0.001 sec)

```
MariaDB [Projects]> select projNo from EmpProject where projNo =  
'SCCS';
```

projNo
SCCS
SCCS

2 rows in set (0.002 sec)

```
MariaDB [Projects]> select count(projNo) from empProject where  
empNo = 'E1';
```

count(projNo)
1

1 row in set (0.001 sec)

```
MariaDB [Projects]> select empNo, totalHours from empProject  
group by empNo;
```

empNo	totalHours
E1	18
E2	12
E3	15
E4	6
E6	20
E7	18

6 rows in set (0.001 sec)

```
MariaDB [Projects]>
```

7.28

The materialised view stores the result of the view in a table. For the ExpensiveParts view, it has to be updated whenever there is a change in the Part table. If a new part is inserted and the partCost greater than 1000, then the partNo should be added to the view. Also, if a partCost is updated from 1000 or below to greater than 1000 then the partNo must also be added.

if partCost moves from greater than 1000 to 1000 or less, the partNo should be removed except if that part does not have another contract where the cost is still higher than 1000. The view is maintained without getting access to the base Part table when the change clearly shows whether the part should be added or kept. When removing a partNo, the base table also needs to be checked to validate that no other contract for that part still costs greater than 1000.